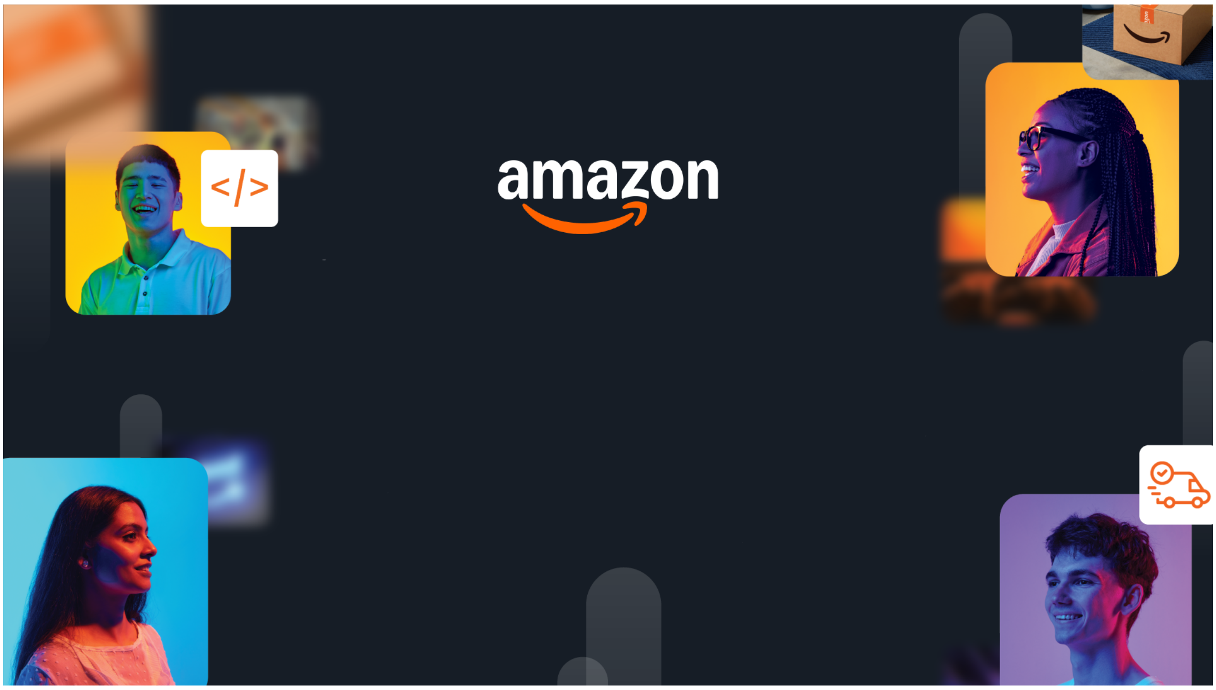

HackOn with Amazon

A Universe of Opportunity

48-Hour Hackathon — Solution Document

Team Name	KKR — Kolkata Kode Riders
Hackathon Theme	Second Life Commerce: AI-Powered Returns & Sustainable Resale
Date	June 15, 2026

Team Members

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1. Problem Statement & Relevance

→ *Jury focus: Innovativeness (novelty, theme alignment) + Degree of Disruption (global relevance)*

The Problem

Every year, over 2.6 billion kg of returned products end up in landfills despite being perfectly usable. Returns cost Amazon $25+ per item in reverse logistics, inspection, and restocking — totaling billions annually. Meanwhile, customers distrust refurbished products, and sellers lack actionable insight into why items are returned.

Why It Matters

- 30% of all online purchases are returned (vs 8% in-store)
- $816 billion worth of merchandise was returned in the US alone in 2022
- Only 5% of returned items are resold at full value — the rest are discarded or destroyed
- Return shipping generates 24 million metric tons of CO_2 annually
- Customers who buy refurbished report 40% lower confidence than buying new

Theme Alignment

Our solution directly addresses "Second Life Commerce" by building an intelligent ecosystem where every returned product automatically finds its optimal next destination — whether that's a nearby buyer, a refurbishment center, a donation partner, or a recycling facility. We don't just process returns; we prevent them before they happen and ensure every item that does come back gets a meaningful second life.

What Makes This Novel

No existing solution connects all three phases: **prevention** (before purchase), **intelligent resolution** (during return), and **circular recovery** (after return) into a single system. Our platform:

1. Prevents returns before they happen through predictive purchase guidance
2. Resolves issues without physical returns when possible (saving logistics costs)
3. Matches returned items to nearby buyers in real-time (demand-aware routing)
4. Builds customer trust in refurbished products through certification + rewards
5. Makes sustainability profitable — not just a cost center

2. Customer & Solution

→ *Jury focus: Quality of Presentation (clarity) + Quality of Implementation (working prototype)*

Target Customer

Three audiences, one connected platform:

- **Customers**: Want to buy confidently, return easily, and trust renewed products
- **Sellers**: Need to understand why products are returned and how to reduce returns
- **Operations**: Need to process returns efficiently and maximize recovery value

How We Solve It

- **Predictive Purchase Prevention** — Before checkout, customers see guidance: "This shoe runs small — check the size guide." Reduces unnecessary returns at the source.
- **Intelligent Resolution Center** — Analyzes item value, shipping cost, and nearby demand to offer optimal resolution: keep-item refund, replacement, or standard return.
- **Demand-Aware Circular Routing** — Returned items matched to active buyers nearby and routed directly — skipping warehouse entirely. Saves $22+ per item.
- **Certified Renewed Marketplace** — Dedicated tab with quality grades, inspection transparency, and 90-day guarantee.
- **Sustainability Rewards** — Customers earn points for buying renewed, choosing keep-item resolutions, and sustainable actions.

User Workflow

```
CUSTOMER: Browse → Guidance → Buy → [Issue?] → Resolution Center
                                          ↓
          Instant Refund | Replace | Return → Drop off → Refund

AMAZON INTERNAL (invisible to customer):
  Return → Fraud Check → Quality Grade → Recovery Decision
                                               ↓
              Resell | Refurbish | Donate | Recycle
                                               ↓
              Demand Matching → Route to Nearest Buyer → Sold
```

Working Prototype

Live deployed across 4 applications:

- Customer App: https://amazon-hackathon-phi.vercel.app/
- Executive Dashboard: https://amazon-hackathon-pjt3.vercel.app/
- Operations Dashboard: https://amazon-hackathon-udpw.vercel.app/
- Seller Dashboard: https://amazon-hackathon-lct1.vercel.app/

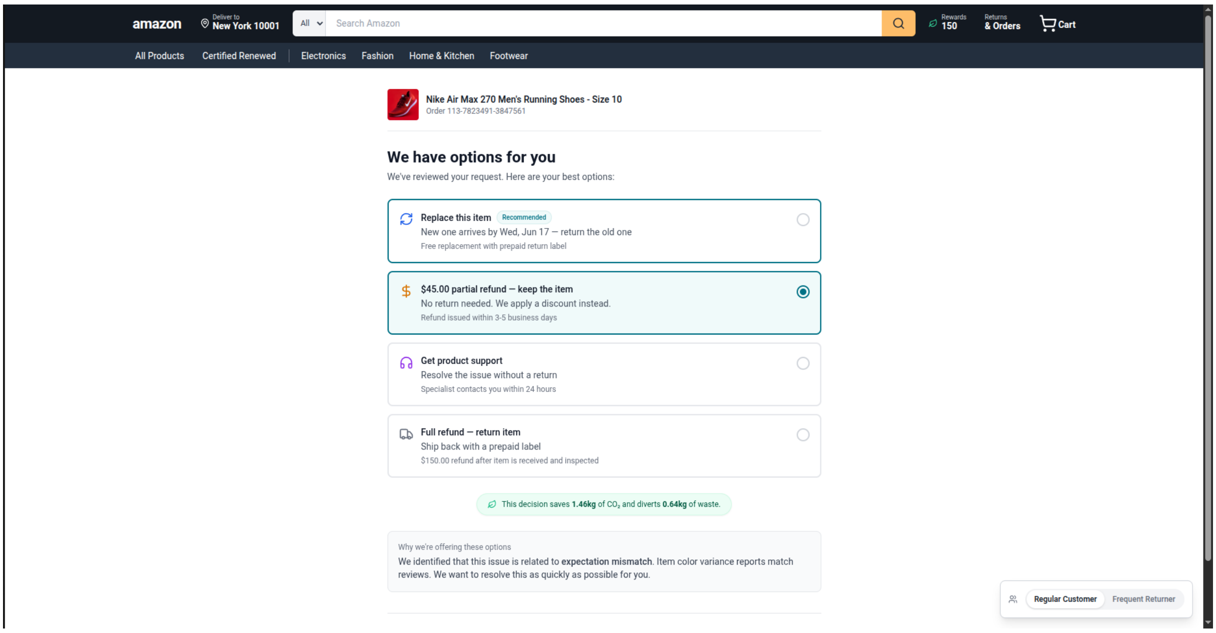
Backend: 12 microservices live on AWS ECS Fargate. All responses are real — no mocked data.

Demo: https://github.com/subh-ghosh/amazon-hackathon

3. Tech Architecture & Scaling

→ *Jury focus: Tech Architecture (complexity, algorithms, APIs) + Scalability (depth, interconnectedness)*

Architecture



Tech Stack

Layer	Technology	Why
Frontend	Next.js 14, TypeScript, Tailwind CSS	SSR proxy routes, type-safe contracts, rapid iteration
Backend	Python 3.10, FastAPI, Pydantic v2	Strict schema enforcement, async I/O, auto OpenAPI
Data/ML	Neptune (Graph), DynamoDB, Bedrock	Knowledge graph for fraud, event sourcing, LLM analysis
Infra	AWS CDK, ECS Fargate, ALB, EventBridge	IaC, zero-server ops, event-driven architecture

Key Algorithms & Complexity

1. **Multi-Factor Recovery Optimizer (S6)** — Weighted decision matrix evaluating 5+ recovery scenarios. Balances recovery value, carbon impact, processing time, fraud risk, and seller trust with penalty/bonus modifiers.
2. **Demand-Aware Circular Routing (S9)** — Constrained optimization matching items to facilities by distance, capacity, type compatibility, and real-time demand signals.
3. **Predictive Return Prevention (S1)** — Composite risk scoring using 9 weighted factors with actionable recommendation generation.
4. **Native Payload Compatibility** — All 12 services designed for zero-transformation pipeline: S5 → S6 → S7 → S9 with direct output-to-input pass-through.

Scaling Strategy

- **Horizontal:** 12 services scale independently via ECS auto-scaling

- **Event-driven:** EventBridge decouples all services. Adding service #13 = zero changes to existing ones
- **Graph intelligence:** Neptune handles fraud network queries in O(1) traversal vs 10+ SQL JOINs
- **Global ready:** CDK deploys entire stack in new region in ¡10 minutes
- **Cost:** Fargate = pay per task. At 1M returns/month: ¡$2,000/month for all 12 services

4. Future Vision

→ *Jury focus: Futuristic Vision (long-term thinking, multi-segment expansion, value impact)*

Where This Goes

In 1–3 years, this becomes Amazon's **Circular Commerce Operating System** — the invisible intelligence layer ensuring no product is ever truly "wasted." Every return becomes an opportunity to resell, refurbish, donate, or learn and prevent the next one.

Roadmap

Horizon	Milestone	Impact
0–3 mo	Renewed marketplace pilot in 3 categories. Prevention on top-100 SKUs.	15% return reduction. 50K renewed items listed.
3–6 mo	Real-time demand matching. Green Credits live. All categories.	$3.2M logistics savings. 200K renewed buyers.
6–12 mo	Seller prevention tools. CV quality grading. Cross-border routing.	25% overall return reduction. 1M items from landfill.

Multi-Segment Expansion

- **Amazon Warehouse Deals** — Powered by certified grading and demand matching
- **Amazon Trade-In** — Devices flow into Renewed pipeline
- **Amazon Global** — Cross-border: returned in US, sold renewed in India
- **Third-party logistics** — License recovery engine to other retailers
- **Amazon Business (B2B)** — Enterprise equipment renewal at scale

Value Impact

At Amazon's scale (500M+ returns/year):

- $12.5B annual savings from prevented returns
- 65M items given second lives instead of landfill
- 4.2M tons CO_2 prevented from unnecessary shipping
- $2.1B new revenue from Renewed marketplace
- 89% customer satisfaction improvement on returns

Links: GitHub: https://github.com/subh-ghosh/amazon-hackathon — Customer: https://amazon-hackathon-phi.vercel.app/ — Executive: https://amazon-hackathon-pjt3.vercel.app/ — Operations: https://amazon-hackathon-udpw.vercel.app/ — Seller: https://amazon-hackathon-lct1.vercel.app/